

Course outline (Syllabus of the course)

Course Code: CS2405 Name of the Programme: B.Tech(Hons.) Name of the Course: Cybersecurity Semester: III		
Course credit	No. of hours per week	Total no. of Teaching hours
03	2+0+2	60
Course Objectives:		
a) To familiarize cybercrime terminologies and perspectives.		
b) Identify and Classify Cyber Threats and Crimes, and measures to enable security.		
c) Explain the risk assessment process and common risk mitigation strategies used in information security.		
d) Gain Hands-on Experience with Cybersecurity Tools.		

Syllabus:

Module - 1 Overview of Information Security and Cybercrime	No. of hours
	7
An Overview of Information Security: Security, CNSS Security model, Components of an Information system, Balancing Information security and access, Threats, Cybercrime and Information security, Classification of Cybercrimes, NIST/ISO frameworks. The legal perspectives- Indian perspective, Global perspective. Types of Attacks, Social Engineering, Cyber stalking, cyberbully, financial frauds.	

Module - 2 Cyber Threats and Security Measures	No. of hours
	6
Phishing, baiting, pretexting, smishing, vishing. Password Cracking, Key-loggers and Spywares, Types of Viruses, Worms, DoS and DDoS, SQL injection, Buffer Overflow, Spyware, Adware and Ransomware. Antivirus. Cyber Kill Chain, MITRE ATT&CK, Threat Intelligence Basics.	

Module - 3 Risk Management and Ethical Practices in Information Security	No. of hours
	5
Overview, Risk identification, Risk Assessment, Risk Control Strategies, Quantitative vs. Qualitative Risk Control Practices. Risk Management. Laws and Ethics in Information Security, Codes of Ethics, Protecting programs and data.	

Module - 4 Security Technology	No. of hours
	6
Access Control, Firewalls, and Intrusion Detection Systems: - Security Technology: Access Control, Firewalls, and VPNs, Intrusion Detection and Prevention Systems, Snort, Honeypots, Honeynets and Padded Cell Systems-overview on scanning and analysis tools.	

Module - 5 Information Gathering and Web Application Security	No. of hours
	6
Information Gathering - Social Engineering, Foot Printing & Scanning, Port Scanners, Nmap, Network scanners, Network Session Analysis, System Integrity Validation. Open Web Application Security Project (OWASP), Web Site Audit and Vulnerabilities assessment.	

Lab Programs		[30 Hours]
1	Install Kali Linux on Virtual box	
2	Explore Kali Linux and bash scripting	
3	Capture network traffic using Wire shark and analyze basic protocols like HTTP, DNS, and SMTP to understand how data is transmitted and received using Wireshark.	
4	Password Sniffing: Simulate a scenario where a password is transmitted in plaintext. Use Wireshark to capture and analyze the packets to demonstrate the vulnerability and the importance of encryption	
5	ARP Poisoning Attack: Set up an ARP poisoning attack using tools like Ettercap. Analyze the captured packets to understand how the attack can lead to a Man-in-the-Middle scenario using Wireshark.	
6	Install Metasploitable2 on the virtual box and search for unpatched vulnerabilities	
7	Use Metasploit to exploit an unpatched vulnerability	
8	Examine real-time network traffic analysis and data packet logging using Snort	
9	Ban IP addresses that show the malicious sign using Fail2ban to scan log files	
10	Launch brute-force attacks on the Linux server using Hydra.	
11	Understand the Nmap command and scan targets using Nmap	
12	Identify the vulnerabilities using OWASP ZAP tool	

Course outcomes
CO1: Understand the foundational concepts of information security and cybercrime, including legal frameworks and threat classification.
CO2: Analyze various cyber threats and explain modern techniques used for threat detection and prevention.
CO3: Apply risk assessment methods and demonstrate understanding of ethical practices and laws in information security.
CO4: Demonstrate the use of security technologies and tools for protecting networks and web applications.

Text books	
1.	Sunit Belapure and Nina Godbole, "Cyber Security: Understanding Cyber Crimes, Computer Forensics and Legal Perspectives", Wiley India Pvt Ltd, ISBN: 978-81-265-21791, 2011, First Edition (Reprinted 2018)
2.	Michael E Whitman and Herbert J Mattord, "Principles of Information Security", Seventh Edition, Cengage Learning, 2021, ISBN: 9780357506561.

3.	William Stallings, “Cryptography and Network Security – Principles and Practices”, Pearson Education; Seventh edition, 2018, ISBN: 9781292158594.
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Reference books	
1.	David Kim, Michael G. Solomon, “Fundamentals of Information Systems Security”, Jones & Bartlett Learning Publishers, 2021, ISBN: 9781284220742
2.	William Stallings, Lawrie Brown, “Computer Security Principles and Practice”, Third Edition, Pearson Education, 2018, ISBN: 9781292220635.
3.	Georgia Weidman, “Penetration Testing: A Hands-On Introduction to Hacking”, No Starch Press, 2014, ISBN: 9781593275648.

Lesson plan

Name of the Programme:	B.Tech(Hons.)	
Title of the Course:	Cybersecurity	
Course code	CS2405	
Semester:	III	
Names of the Course Instructor(s)	Prof. Evlin Vidy Latha P	
Course credit	No. of hours per week (2+0+2)	Total no. of Teaching hours
03	04 Hours	60 Hours

Session	Module No.	Topic	Pedagogy /activities	Pre-reading/reference
1.	1	An Overview of Information Security: Security,	Interactive lecture: PPT	Whitman and Mattord, (Ch 1)
2.		CNSS Security model, Components of an Information system,	Interactive lecture: PPT	Whitman and Mattord, (Ch 1)
3.		Balancing Information security and access,	Interactive lecture: PPT	Whitman and Mattord, (Ch 1)
4.		Threats, Cybercrime and Information security, Classification of Cybercrimes,	Interactive lecture: PPT	Sunit and Nina, (Ch 2)
5.		NIST/ISO frameworks.	Interactive lecture: PPT, case study	Whitman and Mattord, (Ch 6)
6.		The legal perspectives- Indian perspective, Global perspective.	Interactive lecture: PPT	Sunit and Nina, (Ch 8)
7.		Types of Attacks, Social	Interactive lecture: PPT	Sunit and Nina, (Ch 4, 5)



		Engineering, Cyber stalking, cyberbully, financial frauds.		
8.		Lab 1 : Install Kali Linux on Virtual box	Hands-on	Georgia Weidman, (Ch 1)
9.				
10.		Lab 2 : Explore Kali Linux and bash scripting	Hands-on	Georgia Weidman, (Ch 2)
11.				
12.		Phishing, baiting, pretexting, smishing, vishing.	Interactive lecture: PPT	Sunit and Nina, (Ch 6)
13.		Password Cracking, Key-loggers and Spywares,	Interactive lecture: PPT	Stallings, (Ch 21)
14.		Types of Virus, Worms, DoS and DDoS,	Interactive lecture: PPT	Stallings, (Ch 21)
15.		SQL injection, Buffer Overflow,	Hands-on activity	Stallings, (Ch 21)
16.	2	Spyware, Adware and Ransomware. Antivirus. Cyber Kill Chain,	Interactive lecture PPT, Case study	Stallings, (Ch 21), Whitman and Mattord, (Ch 6)
17.		MITRE ATT&CK, Threat Intelligence Basics.	Interactive lecture PPT: Framework explanation	Whitman and Mattord, (Ch 6,9)
18.		Lab 3: Capture and analyze traffic using Wireshark.	Hands-on	Georgia Weidman, (Ch 3)
19.				
20.		Lab 4: Password Sniffing	Hands-on	Georgia Weidman, (Ch 5)
21.				
22.		Lab 5: ARP poisoning	Hands-on	Georgia Weidman, (Ch 5)
23.				
24.		Overview, Risk identification, Risk Assessment,	Interactive lecture PPT	Whitman and Mattord, (Ch 5)
25.	3	Risk Control Strategies,	Interactive lecture PPT, Activity discussion	Whitman and Mattord, (Ch 6)
26.		Quantitative vs. Qualitative Risk Control Practices.	Interactive lecture PPT	Whitman and Mattord, (Ch 6)

		Risk Management.		
27.		Laws and Ethics in Information Security,	Interactive lecture PPT	Sunit and Nina, (Ch 8)
28.		Codes of Ethics, Protecting programs and data.	Interactive lecture PPT	Sunit and Nina, (Ch 8)
29.		Lab 6: search for unpatched vulnerabilities using Metasploitable 2	Hands-on	Georgia Weidman, (Ch 6)
30.				
31.		Lab 7: Use Metasploit to exploit an unpatched vulnerability	Hands-on	Georgia Weidman, (Ch 6)
32.				
33.		Security Technology: Access Control,	Interactive lecture PPT	Stallings, (Ch 14)
34.		Firewalls,	Interactive lecture and Hands-on Demonstration	Stallings, (Ch 14)
35.		VPNs,	Interactive lecture and Hands-on Demonstration	Stallings, (Ch 14)
36.		Intrusion Detection and Prevention Systems,	Interactive lecture PPT	Stallings, (Ch 14)
37.	4	Snort, Honeypots, Honeynets, and Padded Cell Systems	Interactive lecture PPT	Georgia Weidman, (Ch 5, 6)
38.		overview of scanning and analysis tools.	Interactive lecture and Hands-on Demonstration	Georgia Weidman, (Ch 5, 6)
39.		Lab 8: Examine real-time network traffic analysis and data packet logging using Snort	Hands-on	Georgia Weidman, (Ch 5)
40.				



41.		Lab 9: Ban IP addresses that show the malicious sign using Fail2ban to scan log files	Hands-on	Georgia Weidman, (Ch 7)
42.				
43.		CIE 3 Case Study discussion	Student Presentation	
44.				
45.		Information Gathering - Social Engineering, Foot Printing	Interactive lecture PPT	Georgia Weidman, (Ch 4)
46.		Scanning, Port Scanners,	Interactive lecture PPT, Hands-on	Georgia Weidman, (Ch 6,7)
47.		Nmap,	Interactive lecture PPT, Hands-on	Georgia Weidman, (Ch 7)
48.		Network scanners,	Interactive lecture PPT	Georgia Weidman, (Ch 7)
49.		Network Session Analysis, System Integrity Validation.	Interactive lecture PPT, Hands-on	Georgia Weidman, (Ch 7,8)
50.	5	Open Web Application Security Project (OWASP), Web Site Audit and Vulnerabilities assessment.	Interactive lecture PPT, Hands-on	Georgia Weidman, (Ch 8)
51.		Lab 10: Launch brute-force attacks on the Linux server using Hydra.	Hands-on	Georgia Weidman, (Ch 7)
52.				
53.		Lab11: Understand the Nmap command and scan targets using Nmap	Hands-on	Georgia Weidman, (Ch 7)
54.				
55.		Lab12: Identify the vulnerabilities	Hands-on	Georgia Weidman, (Ch 8)
56.				

		using OWASP ZAP tool		
57.		Role Play activity on social engineering simulation	Roleplay Activity	
58.				
59.		CASE STUDY REVIEW	STUDENT PRESENTATION	
60.				

Assessment and Evaluation

Continuous Internal Evaluation (CIE) :70%	
Components of CIE	Weightage of each component
CIE-1 - Quiz Graded Component 1 (based on Theory concepts and IBM Skills Build Course) - IBM Skills Build Courses: Graded Component-2	15 marks out of 70 5 marks out of 70
CIE-2 - Theory	25 marks out of 70
CIE-3 - Lab & viva - Case study review Submission	10 marks out of 70 15 marks out of 70

Semester End Exams (SEE): 30%	
Mode of Exam	Theory
Weightage of exam	30%

Course Outcomes- Programme Outcomes Matrix

Program Outcome Course Outcome		PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	Understand the foundational	3	2		2						2		

	concepts of information security and cybercrime, including legal frameworks and threat classification.												
CO2	Analyse various cyber threats and explain modern techniques used for threat detection and prevention.	2	3		2			2	2		2		
CO3	Apply risk assessment methods and demonstrate understanding of ethical practices and laws in information security.	2	3	2				2	2		3		
CO4	Demonstrate the use of security technologies and tools for protecting networks and web applications.	3	2	2	3			2	2		1		

Rubrics for evaluation of CIE (separate for each CIE component)

Total Marks		20
CIE-1 Components		Rubrics for Assessment
CIE-1	Quiz Graded Component 1	The quiz will be conducted for 15 marks. Each right answer carries 0.5 mark each. No negative marking for wrong answers.
	IBM Skill Build Courses: Graded Component-2	Certificate of completion for 5 marks. The specified courses on the IBM Skill Build platform completion certificate to be uploaded to the designated Google Classroom. <i>(Refer Table-1)</i>

Table-1:

Graded Component-2 (5 Marks):

Assess students' ability to complete assigned courses on the IBM Skill Build platform and reflect on their learning.

Criteria	Excellent (5 Marks)	Satisfactory (3-4 Marks)	Needs Improvement (0-2 Marks)
Completion of Courses (5 Marks)	Completes all specified courses on the IBM Skill Build platform before the deadline. Certificates are uploaded to Google Classroom with no errors.	Completes some courses or has minor errors in uploading certificates.	Fails to complete the courses or does not upload certificates.

Total Marks		25
CIE-2 Components		Rubrics for Assessment
CIE-2	Theory	The test will be conducted for 25 marks and will be evaluated according to the Scheme of Evaluation prepared by the course faculty.

Total Marks		25
CIE-3 Components		Rubrics for Assessment
CIE-3	Practical Component	Lab + Viva = 10 marks (<i>Refer Table-2</i>) Case Study Review =15 marks (<i>Refer Table-3</i>)

Table-2

Lab Execution rubrics (Max: 6 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Understanding of problem statements and preparedness (2 Marks)	Completeness of lab setup (e.g., VM, tools pre-installed)	Students exhibit a thorough understanding of the requirements and are fully prepared with all tools and setup ready	Students have sufficient knowledge of the requirements and are partially prepared with installation.	Students exhibit little understanding of requirements, and are poorly prepared (multiple missing setups)
2.	Execution (2 Marks)	Running commands, executing tasks, and capturing results	Executes all steps correctly and efficiently	Executes with minor issues or guidance	Unable to execute key tasks independently
3.	Results and Documentation (2 Marks)	Screenshots, logs, steps, and conclusions in the lab report	Well-documented with clear evidence and interpretation	Incomplete or unclear documentation	Minimal documentation, hard to follow
Viva Voce Rubrics (Max: 4 marks)					
Sl. No	Criteria	Measuring Methods	Excellent (2 Marks)	Good (1 Mark)	Poor (0.5 Marks)
1.	Conceptual Understanding (2 Marks)	Viva Voce	Thoroughly explains the related concepts	Adequately explains the related concepts	Unable to explain related concepts

2.	Problem solving ability (2 Mark)	Viva Voce	Demonstrates clear, logical approach to solving problems, with reasoning	Identifies issues and partially solves problems	Struggles to solve problems
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Table-3

Grading Rubrics for Case study (Total: 15 Marks)					
Sl. No	Criteria	Excellent	Very Good	Good	Needs Improvement
1.	Relevance, Background & Context (4 Marks)	A case study is highly relevant to cybersecurity, clearly explains context, timeline, affected systems, and key stakeholders. (4 M)	The case study is relevant, with a clear background, but it may lack some context. (3 M)	The case study is relevant; the background is clear but lacks depth and context. (2 M)	The case study is only somewhat relevant or poorly justified; a limited or vague background is provided. (0-1 M)
2.	Technical, Legal & Ethical Analysis (4 Marks)	Strong technical analysis of attack/tools used; accurately covers vulnerabilities, impact, legal issues, and ethical dimensions. (4 M)	Moderate technical analysis; touches on tools and laws but lacks detail or critical depth.	Moderate technical analysis; touches on tools and laws but lacks detail or critical depth.	Minimal or incorrect technical content; ethical/legal aspects poorly addressed or missing. (0 M)
3.	Insight, Originality & Presentation (4 Marks)	Demonstrates critical thinking and original insights; well-structured, visually clear, and professionally formatted. (3 M)	Some insights present; structure is clear, but presentation could be improved. (2 M)	Basic feedback is provided but lacks depth or critical analysis. (1 M)	Lacks original thought or clear structure; poor formatting or presentation. (0 M)

4.	Report Documentation (3 Marks)	Comprehensive, well-organized, and insightful report that covers all aspects in depth. (3 M)	Thorough report with minor areas for improvement. 2 M)	The report covers all essentials but lacks detail or organization. (1.5 M)	Incomplete or significantly flawed report. (0-1 M)
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Signature of the Course Faculty